Python Loops Masterclass

100 For & While Loop Questions with Complete Solutions

Basic se lekar Advanced logic building tak ke 100 chune hue sawaal
aur unke efficient Python codes.

Structured Practice Guide

Levels 1 - 6 • Complete Reference Sheet

Level 1: Basic For Loop (Questions 1-20)

Q1. 1 se 10 tak ke saare numbers print karein.

```
for i in range(1, 11):
    print(i)
```

Q2. 10 se 1 tak ke saare numbers reverse order mein print karein.

```
for i in range(10, 0, -1):
    print(i)
```

Q3. Ek string "PYTHON" ke har ek character ko alag line mein print karein.

```
for char in "PYTHON":
    print(char)
```

Q4. Pehle 10 even (sam) numbers print karein.

```
for i in range(1, 11):
    print(i * 2)
```

Q5. Pehle 10 odd (visham) numbers print karein.

```
for i in range(1, 11):
    print(i * 2 - 1)
```

Q6. 1 se 20 tak ke saare numbers ka sum (jod) calculate karein.

```
total_sum = 0
for i in range(1, 21):
    total_sum += i
print("Sum:", total_sum)
```

Q7. User se ek number lein aur uska 1 se 10 tak ka table (pahada) print karein.

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

Q8. Ek list [10, 20, 30, 40, 50] ke saare elements ko bari-bari print karein.

```
my_list = [10, 20, 30, 40, 50]
for element in my_list:
    print(element)
```

Q9. 1 se 50 ke beech ke sirf woh numbers print karein jo 5 se divide hote hain.

```
for i in range(1, 51):
    if i % 5 == 0:
        print(i)
```

Q10. Pehle 10 natural numbers ka square (varg) print karein.

```
for i in range(1, 11):
    print(f"Square of {i} is {i ** 2}")
```

Q11. Pehle 10 natural numbers ka cube (ghan) print karein.

```
for i in range(1, 11):
    print(f"Cube of {i} is {i ** 3}")
```

Q12. Ek list ['apple', 'banana', 'cherry'] ke har item ke sath "I like" jodkar print karein.

```
fruits = ['apple', 'banana', 'cherry']
for fruit in fruits:
    print("I like", fruit)
```

Q13. 1 se 100 ke beech ke saare even numbers ka sum nikaalein.

```
even_sum = 0
for i in range(1, 101):
    if i % 2 == 0:
        even_sum += i
print("Sum of even numbers:", even_sum)
```

Q14. 1 se 100 ke beech ke saare odd numbers ka sum nikaalein.

```
odd_sum = 0
for i in range(1, 101):
    if i % 2 != 0:
        odd_sum += i
print("Sum of odd numbers:", odd_sum)
```

Q15. Ek string mein kitne characters hain, count karein (bina len() function use kiye).

```
my_string = "Hello Python"
count = 0
for char in my_string:
    count += 1
print("Length is:", count)
```

Q16. 1 se 15 tak ke saare numbers ka factorial nikaalein.

```
fact = 1
for i in range(1, 16):
    fact *= i
    print(f"Factorial of {i} is {fact}")
```

Q17. Ek list mein se sirf negative numbers ko filter karke print karein.

```
numbers = [10, -5, 23, -8, 0, -1, 4]
for num in numbers:
    if num < 0:
        print(num)
```

Q18. range(5, 50, 5) ka use karke output print karein aur dekhein kya aata hai.

```
for i in range(5, 50, 5):
    print(i)
```

Q19. Ek list ke saare numbers ko double karke ek nayi list banayein.

```
old_list = [1, 2, 3, 4, 5]
new_list = []
for num in old_list:
    new_list.append(num * 2)
print("New List:", new_list)
```

Q20. Apna naam 5 baar print karein.

```
for i in range(5):
    print("Aapka Naam")
```

Level 2: Basic While Loop (Questions 21-40)

Q21. while loop ka use karke 1 se 10 tak numbers print karein.

```
i = 1
while i <= 10:
    print(i)
    i += 1
```

Q22. while loop se 10 se 1 tak ulti ginti (reverse counting) print karein.

```
i = 10
while i >= 1:
    print(i)
    i -= 1
```

Q23. Jab tak user 0 enter nahi karta, tab tak user se input lete rahein.

```
num = -1
while num != 0:
    num = int(input("Enter 0 to stop: "))
```

Q24. 1 se 10 tak ke numbers ka sum nikaalein while loop se.

```
i = 1
total_sum = 0
while i <= 10:
    total_sum += i
    i += 1
print("Sum:", total_sum)
```

Q25. Ek number ka table print karein while loop ka use karke.

```
num = int(input("Enter number for table: "))
i = 1
while i <= 10:
    print(f"{num} x {i} = {num * i}")
    i += 1
```

Q26. 1 se 20 ke beech ke saare odd numbers print karein.

```
i = 1
while i <= 20:
    if i % 2 != 0:
        print(i)
    i += 1
```

Q27. Ek variable x = 1 lein aur use tab tak double karte rahein jab tak woh 1000 se bada na ho jaye. Count karein kitni baar loop chala.

```
x = 1
count = 0
while x <= 1000:
    x = x * 2
    count += 1
print("Final value:", x)
print("Loop counts:", count)
```

Q28. User se ek password input karwayein. Jab tak password "secret123" na ho, tab tak "Wrong password, try again" bolte rahein.

```
password = ""
while password != "secret123":
    password = input("Enter password: ")
    if password != "secret123":
        print("Wrong password, try again.")
print("Access Granted!")
```

Q29. while loop se ek string ko reverse order mein print karein.

```
my_str = "Python"
index = len(my_str) - 1
while index >= 0:
    print(my_str[index], end="")
    index -= 1
print()
```

Q30. Pehle 10 numbers ka square print karein while loop se.

```
i = 1
while i <= 10:
    print(f"Square of {i} is {i**2}")
    i += 1
```

Q31. Ek counter banayein jo 10 se 100 tak 10-10 ke gap mein print kare (10, 20, 30...).

```
counter = 10
while counter <= 100:
    print(counter)
    counter += 10
```

Q32. Ek number ka factorial nikaalein while loop se.

```
num = int(input("Enter number: "))
fact = 1
i = num
while i > 0:
    fact *= i
    i -= 1
print("Factorial:", fact)
```

Q33. Infinite loop banayein aur use break se rokna seekhein.

```
while True:
    print("Yeh ek baar chalega.")
    break  # Break statement infinite loop ko rok deta hai
```

Q34. User se numbers lete rahein aur unka sum karte rahein. Jaise hi user koi negative number daale, loop rok dein.

```
total = 0
while True:
    num = int(input("Enter a number (negative to stop): "))
    if num < 0:
        break
    total += num
print("Total Sum:", total)
```

Q35. while loop ka use karke ek list ke saare elements ko pop (remove) karein jab tak list khali na ho jaye.

```
my_list = [1, 2, 3, 4, 5]
while len(my_list) > 0:
    removed = my_list.pop()
    print("Removed:", removed, "Remaining:", my_list)
```

Q36. 1 se 30 ke beech ke saare numbers jo 3 aur 5 dono se divide hote hain, unhein print karein.

```
i = 1
while i <= 30:
    if i % 3 == 0 and i % 5 == 0:
        print(i)
    i += 1
```

Q37. Ek number ke saare digits ka sum nikaalein (e.g., 123 ka output 1+2+3 = 6).

```
num = 123
digit_sum = 0
while num > 0:
    digit = num % 10
    digit_sum += digit
    num = num // 10
print("Sum of digits:", digit_sum)
```

Q38. Ek number ko reverse karein (e.g., input: 1234, output: 4321).

```
num = 1234
reversed_num = 0
while num > 0:
    digit = num % 10
    reversed_num = (reversed_num * 10) + digit
    num = num // 10
print("Reversed:", reversed_num)
```

Q39. Check karein ki koi number Palindrome hai ya nahi (e.g., 121).

```
original_num = 121
num = original_num
reversed_num = 0
while num > 0:
    digit = num % 10
    reversed_num = (reversed_num * 10) + digit
    num = num // 10
if original_num == reversed_num:
    print("Yes, it's a Palindrome")
else:
    print("No, it's not")
```

Q40. User se 'yes' ya 'no' input lein. Jab tak galat input ho, dobara puchein.

```
user_input = ""
while user_input not in ["yes", "no"]:
    user_input = input("Please enter 'yes' or 'no': ").lower()
print("Thank you! You entered:", user_input)
```

Level 3: Break, Continue, aur Pass (Questions 41-55)

Q41. 1 se 10 tak loop chalayein, lekin agar number 5 aaye toh loop ko break kar dein.

```
for i in range(1, 11):
    if i == 5:
        break
    print(i)
```

Q42. 1 se 10 tak loop chalayein, lekin number 5 ko continue ka use karke skip karein.

```
for i in range(1, 11):
    if i == 5:
        continue
    print(i)
```

Q43. Ek string ke characters print karein, jaise hi koi space (' ') aaye, loop rok dein.

```
text = "Hello World"
for char in text:
    if char == " ":
        break
    print(char, end="")
print()
```

Q44. 1 se 50 tak ke numbers print karein, lekin jo 3 se divide hote hain unhein skip kar dein.

```
for i in range(1, 51):
    if i % 3 == 0:
        continue
    print(i)
```

Q45. Ek infinite while True loop banayein aur usmein user se "exit" likhne par break karein.

```
while True:
    cmd = input("Type 'exit' to stop: ")
    if cmd == "exit":
        break
```

Q46. Ek list mein se numbers print karein, agar koi number 100 se bada mile toh loop wahin khatam kar dein.

```
nums = [10, 45, 89, 120, 30, 50]
for n in nums:
    if n > 100:
        print(f"Found {n}, stopping.")
        break
    print(n)
```

Q47. Ek loop banayein jo sirf list ke string items ko print kare aur numbers ko continue kar de.

```
mixed_list = ["apple", 12, "banana", 45, "cherry"]
for item in mixed_list:
    if type(item) != str:
        continue
    print(item)
```

Q48. pass statement ka use karke ek empty for loop banayein jo error na de.

```
for i in range(5):
    pass  # Future code ke liye placeholder
```

Q49. User se 5 positive numbers lein. Agar user negative number daale toh continue karein.

```
count = 0
while count < 5:
    num = int(input("Enter positive number: "))
    if num < 0:
        print("Negative ignored.")
        continue
    count += 1
```

Q50. Ek list mein prime number dhoondhein, milte hi break karein aur uska index print karein.

```
nums = [4, 6, 8, 9, 11, 14, 15]
for index, num in enumerate(nums):
    is_prime = True
    if num > 1:
        for i in range(2, num):
            if num % i == 0:
                is_prime = False
                break
        if is_prime:
            print(f"First prime {num} found at index {index}")
            break
```

Q51. 1 se 20 tak ke numbers mein agar number even hai toh uska square print karein, odd hai toh skip karein.

```
for i in range(1, 21):
    if i % 2 != 0:
        continue
    print(f"Square of even {i} is {i**2}")
```

Q52. Ek string mein se saare vowels (a, e, i, o, u) ko skip karke baaki characters print karein.

```
text = "programming"
vowels = "aeiou"
for char in text:
    if char in vowels:
        continue
    print(char, end="")
print()
```

Q53. Ek loop chalayein jo 1 se 100 tak jaye, lekin jaise hi pehla number mile jo 11 se divide hota hai, use print karke break ho jaye.

```
for i in range(1, 101):
    if i % 11 == 0:
        print("First number divisible by 11 is:", i)
        break
```

Q54. Ek menu-driven program banayein (1. Start, 2. Options, 3. Exit).

```
while True:
    print("\n1. Start\n2. Options\n3. Exit")
    choice = input("Enter choice: ")
    if choice == "3":
        print("Exiting...")
        break
    elif choice == "1":
        print("Starting Game...")
    elif choice == "2":
        print("Showing Options...")
    else:
        print("Invalid Choice!")
```

Q55. for loop ke sath else block ka use karein jo loop successfully complete hone par message print kare.

```
for i in range(3):
    print("Looping...")
else:
    print("Loop completed successfully without break!")
```

Level 4: Nested Loops & Patterns (Questions 56-75)

Q56. Square Pattern print karein.

```
for i in range(3):
    for j in range(4):
        print("*", end=" ")
    print()
```

Q57. Right Triangle Pattern print karein.

```
for i in range(1, 5):
    for j in range(i):
        print("*", end=" ")
    print()
```

Q58. Inverted Right Triangle Pattern print karein.

```
for i in range(4, 0, -1):
    for j in range(i):
        print("*", end=" ")
    print()
```

Q59. Number Pattern 1 print karein.

```
for i in range(1, 4):
    for j in range(i):
        print(i, end=" ")
    print()
```

Q60. Number Pattern 2 print karein.

```
for i in range(1, 4):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

Q61. Floyd's Triangle print karein.

```
num = 1
for i in range(1, 4):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

Q62. 1 se 5 tak ke saare tables nested loop se print karein.

```
for i in range(1, 6):
    print(f"Table of {i}:", end=" ")
    for j in range(1, 11):
        print(i * j, end=" ")
    print()
```

Q63. Matrix [[1, 2], [3, 4], [5, 6]] ke saare numbers ko print karein.

```
matrix = [[1, 2], [3, 4], [5, 6]]
for row in matrix:
    for item in row:
        print(item, end=" ")
    print()
```

Q64. Pyramid Pattern print karein.

```
n = 3
for i in range(n):
    print(" " * (n - i - 1), end="")
    print("*" * (2 * i + 1))
```

Q65. List mein duplicates hain ya nahi, nested loop se check karein.

```
items = [1, 2, 3, 4, 2, 5]
duplicate = False
for i in range(len(items)):
    for j in range(i + 1, len(items)):
        if items[i] == items[j]:
            duplicate = True
            break
print("Contains Duplicate:", duplicate)
```

Q66. Do lists ke saare possible combinations print karein.

```
l1 = ['Kuch', 'Naya']
l2 = ['Khao', 'Sikho']
for x in l1:
    for y in l2:
        print(x, y)
```

Q67. Alphabet Pattern print karein.

```
for i in range(3):
    char = chr(65 + i) # 65 is ASCII for 'A'
    for j in range(i + 1):
        print(char, end=" ")
    print()
```

Q68. Diamond Pattern print karein.

```
n = 3
for i in range(n):
    print(" " * (n - i - 1) + "*" * (2 * i + 1))
for i in range(n - 2, -1, -1):
    print(" " * (n - i - 1) + "*" * (2 * i + 1))
```

Q69. 3x3 grid indices print karein.

```
for row in range(3):
    for col in range(3):
        print(f"({row},{col})", end=" ")
    print()
```

Q70. Chessboard jaisa pattern (0 aur 1 alternating).

```
for i in range(4):
    for j in range(4):
        print((i + j) % 2, end=" ")
    print()
```

Q71. Kisi string ke saare possible substrings print karein.

```
s = "ABC"
for i in range(len(s)):
    for j in range(i + 1, len(s) + 1):
        print(s[i:j])
```

Q72. Pehle 5 numbers ke factorials ki ek list banayein nested loop se.

```
fact_list = []
for i in range(1, 6):
    fact = 1
    for j in range(1, i + 1):
        fact *= j
    fact_list.append(fact)
print(fact_list)
```

Q73. Inverted pyramid pattern banayein.

```
n = 3
for i in range(n, 0, -1):
    print(" " * (n - i) + "*" * (2 * i - 1))
```

Q74. Reverse number row pattern banayein.

```
for i in range(3, 0, -1):
    for j in range(i, 0, -1):
        print(j, end=" ")
    print()
```

Q75. Bubble Sort algorithm nested loop se design karein.

```
arr = [64, 34, 25, 12, 22, 11, 90]
n = len(arr)
for i in range(n):
    for j in range(0, n - i - 1):
        if arr[j] > arr[j + 1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]
print("Sorted Array:", arr)
```

Level 5: Logic Building & Real Problems (Questions 76-90)

Q76. Fibonacci Series ke pehle 10 numbers print karein.

```
a, b = 0, 1
for _ in range(10):
    print(a, end=" ")
    a, b = b, a + b
print()
```

Q77. Check karein ki di gayi sankhya Prime (Abhajya) hai ya nahi.

```
num = 29
is_prime = True
if num <= 1:
    is_prime = False
else:
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
print("Is Prime:", is_prime)
```

Q78. 1 se 100 ke beech ke saare Prime Numbers print karein.

```
for num in range(2, 101):
    is_prime = True
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print(num, end=" ")
print()
```

Q79. FizzBuzz Problem: 1 se 50 tak numbers print karein (Divisible by 3 -> Fizz, 5 -> Buzz, both -> FizzBuzz).

```
for i in range(1, 51):
    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

Q80. Ek list mein sabse bada (Maximum) number dhoondhein (bina max() use kiye).

```
nums = [4, 7, 1, 9, 2, 8]
max_num = nums[0]
for n in nums:
    if n > max_num:
        max_num = n
print("Maximum:", max_num)
```

Q81. Ek list mein sabse chota (Minimum) number dhoondhein (bina min() use kiye).

```
nums = [4, 7, 1, 9, 2, 8]
min_num = nums[0]
for n in nums:
    if n < min_num:
        min_num = n
print("Minimum:", min_num)
```

Q82. Ek list ko poori tarah reverse karein (bina reverse() ya slicing ke).

```
arr = [1, 2, 3, 4, 5]
n = len(arr)
for i in range(n // 2):
    arr[i], arr[n - i - 1] = arr[n - i - 1], arr[i]
print("Reversed list:", arr)
```

Q83. Check karein ki koi number Armstrong Number hai ya nahi (e.g., 153).

```
num = 153
temp = num
sum_pow = 0
n = len(str(num))
while temp > 0:
    digit = temp % 10
    sum_pow += digit ** n
    temp //= 10
print("Is Armstrong:", num == sum_pow)
```

Q84. Ek list mein har ek element ki frequency count karein loop se.

```
arr = [1, 2, 2, 3, 1, 4, 2]
freq = {}
for item in arr:
    if item in freq:
        freq[item] += 1
    else:
        freq[item] = 1
print("Frequency:", freq)
```

Q85. Do sorted lists ko bina sort() function ke merge karke ek sorted list banayein.

```
l1 = [1, 3, 5]
l2 = [2, 4, 6]
merged = []
i, j = 0, 0
while i < len(l1) and j < len(l2):
    if l1[i] < l2[j]:
        merged.append(l1[i])
        i += 1
    else:
        merged.append(l2[j])
        j += 1
merged.extend(l1[i:])
merged.extend(l2[j:])
print("Merged Sorted List:", merged)
```

Q86. Ek sentence mein sabse bada word loop se pata karein.

```
sentence = "Python is an amazing language"
words = sentence.split()
longest_word = ""
for word in words:
    if len(word) > len(longest_word):
        longest_word = word
print("Longest Word:", longest_word)
```

Q87. Kisi number ka GCD ya HCF nikaalein loop se.

```
a, b = 24, 36
while b:
    a, b = b, a % b
print("GCD:", a)
```

Q88. Kisi number ka LCM nikaalein loop se.

```
a, b = 12, 18
max_num = max(a, b)
while True:
    if max_num % a == 0 and max_num % b == 0:
        lcm = max_num
        break
    max_num += 1
print("LCM:", lcm)
```

Q89. Ek list mein se saare duplicate elements hata kar sirf unique elements ki list banayein.

```
arr = [1, 2, 2, 3, 4, 4, 5]
unique = []
for x in arr:
    if x not in unique:
        unique.append(x)
print("Unique List:", unique)
```

Q90. User se ek binary number lein aur use loop ke zariye Decimal number mein badlein.

```
binary = "1010"
decimal = 0
for digit in binary:
    decimal = decimal * 2 + int(digit)
print("Decimal:", decimal)
```

Level 6: Dictionary & Advance Loop Handling (Questions 91-100)

Q91. Ek dictionary {'a': 1, 'b': 2, 'c': 3} ke saare keys aur values ko loop se print karein.

```
my_dict = {'a': 1, 'b': 2, 'c': 3}
for key, val in my_dict.items():
    print(f"Key: {key}, Value: {val}")
```

Q92. List Comprehension ka use karke 1 se 10 tak ke squares ki ek list banayein.

```
squares = [x**2 for x in range(1, 11)]
print(squares)
```

Q93. enumerate() function ka use karke list ke items ko unke index ke sath print karein.

```
fruits = ['apple', 'banana', 'cherry']
for index, fruit in enumerate(fruits):
    print(f"Index {index}: {fruit}")
```

Q94. Do lists ko ek sath loop mein chalayein zip() function ka use karke.

```
names = ["Amit", "Rahul", "Vijay"]
scores = [85, 92, 78]
for name, score in zip(names, scores):
    print(f"{name} scored {score}")
```

Q95. Ek dictionary mein se sirf un keys ko print karein jinki value 50 se zyada ho.

```
scores = {'Math': 90, 'Science': 45, 'English': 72}
for subject, score in scores.items():
    if score > 50:
        print(subject)
```

Q96. Ek string di gayi hai, loop se har character ka frequency count dictionary mein store karein.

```
text = "hello"
char_count = {}
for char in text:
    char_count[char] = char_count.get(char, 0) + 1
print(char_count)
```

Q97. Dictionary Comprehension ka use karke ek dictionary banayein jahan keys 1 se 5 ho aur values unka cube.

```
cubes_dict = {x: x**3 for x in range(1, 6)}
print(cubes_dict)
```

Q98. Ek nested dictionary ke saare data ko clean format mein loop chalakar print karein.

```
students = {
    'stud1': {'name': 'Aman', 'grade': 'A'},
    'stud2': {'name': 'Rohit', 'grade': 'B'}
}
for s_id, info in students.items():
    print(f"ID: {s_id} -> Name: {info['name']}, Grade: {info['grade']}")
```

Q99. Ek list ke andar jo sub-lists hain un sabko jodkar ek single list banayein (Flatten a list).

```
nested_list = [[1, 2], [3, 4], [5, 6]]
flat_list = []
for sublist in nested_list:
    for item in sublist:
        flat_list.append(item)
print(flat_list)
```

Q100. Ek list comprehension banayein jo 1 se 20 tak ke sirf even numbers ko select kare.

```
even_nums = [x for x in range(1, 21) if x % 2 == 0]
print(even_nums)
```